



Role of secondary shear-layer vortices in the development of flow asymmetry on a cone–cylinder body at high angles of incidence

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Abstract

Vortex asymmetry develops on slender bodies at high angles of incidence, leading to large side-forces and yaw moments even in symmetrical flight conditions. In the present experimental study, the flow over a slender body consisting of a conical forebody and a cylindrical aftbody is studied at high angles of incidence at low speeds. Experiments consisted of force measurements using a six-component strain gauge balance and velocity field using Particle Image Velocimetry. Results show that a pair of counter-rotating vortices develop nearly symmetrically over a polished cone but become asymmetric over the cylinder. Several secondary shear-layer vortices develop at the cone–cylinder junction and over the cylinder due to sharp turning angle, surface imperfections, and joints. These secondary shear-layer vortices then merge with the primary vortex pair, leading to the flow asymmetry. The flow characteristics and side-forces vary significantly when the conical forebody is rotated about its axis of symmetry and set to different azimuthal positions. The results strongly suggest that a change in the geometry at the cone–cylinder junction and surface imperfections present on the cylindrical aftbody play an important role in determining the overall flow-asymmetry and the magnitude of side-forces generated on the slender body at high angles of incidence.

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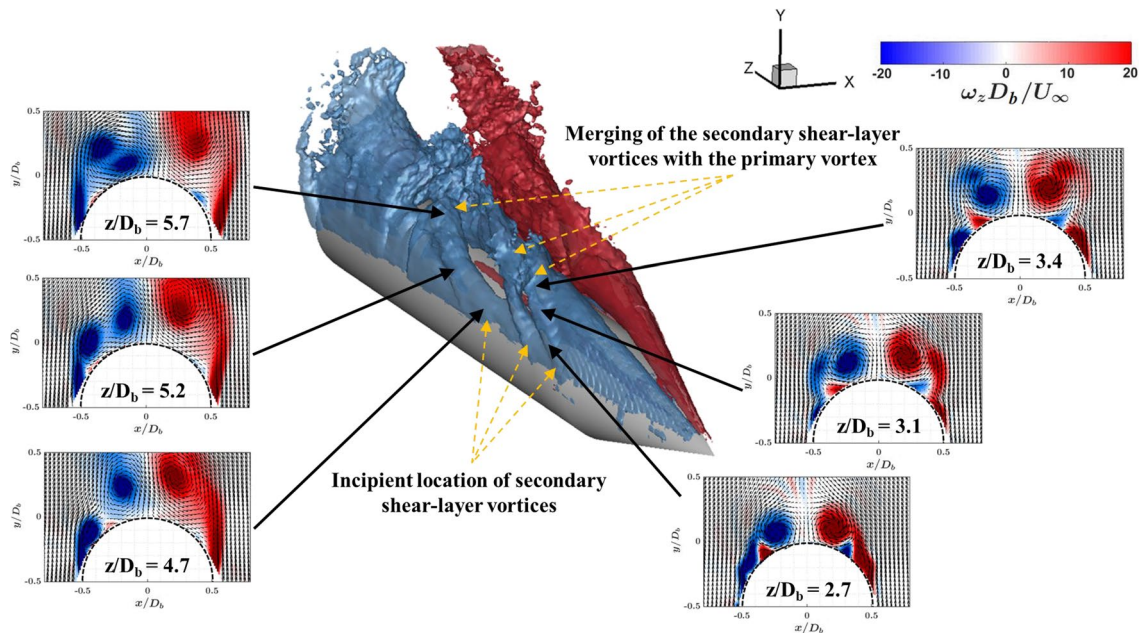
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Graphic abstract

Role of Secondary Shear-Layer Vortices in the Development of Flow Asymmetry on a Cone-Cylinder Body at High Angles of Incidence



1 Introduction

A slender body of revolution like a smart projectile develops large side forces and yaw moments due to vortex asymmetry at high angles of incidence. The problem is predominant at low-to-moderate subsonic speeds where high alpha maneuvers are generally performed. The vortex dominated asymmetric flow interacts with control surfaces, thereby reducing their effectiveness and further aggravating the problem. The generic flow over a slender body can be classified based on the angle of incidence. At low angles of incidence ($\alpha < 12^\circ$), the flow is axially dominated and completely attached to the surface. At moderate angles of incidence ($12^\circ < \alpha < 25^\circ$), the cross-flow separates leading to the formation of a pair of symmetric counter-rotating line vortices. A vortex asymmetry develops as the angle of incidence equals the apex angle of the cone and grows in prominence along the cylindrical body with a further increase in incidence, Fig. 1. This vortex asymmetry leads to a net side-force and yaw-moment on the body. At very high angles of incidence ($\alpha > 50^\circ$), the flow is similar to that of a bluff body resulting in classical, time-dependent Karman vortex shedding.

The generated side-force is a function of flow properties and geometric parameters. Flow properties that significantly influence the magnitude of side force are the Reynolds number and Mach number. Geometric parameters include model surface conditions, nose apex angle, the cross-sectional shape of the body, slenderness ratio (Keener and Chapman 1974), and the nose roll orientation. A numerical study on a spinning and non-spinning projectile carried out by DeSpirito (2017) shows the vortex switching and its effect on the side force magnitude and direction at high angles of incidence. Previous studies

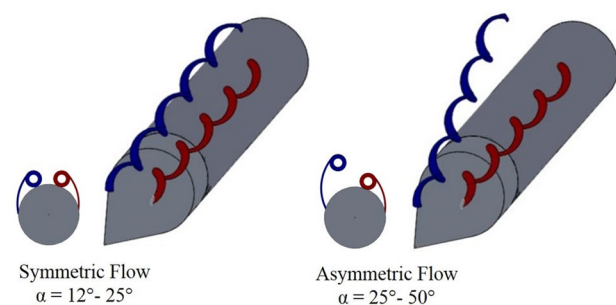


Fig. 1 Schematic of vortex development over a long slender body

(Hunt 1982; Champigny 1994; Kumar et al. 2005, 2008; Ericsson and Reding 1992; Williams 1982; Degani and Tobak 1992; Fagley and McLaughlin 2014) have indicated that the main cause of vortex asymmetry are random micro-surface imperfections. These imperfections are typically of the order of cross-stream boundary layer thickness near the tip of the pointed forebody. This leads to an asymmetric separation of the boundary layer and an asymmetric vortex pair formation (Taligoski et al. 2015). Typically, one of the vortices of the asymmetric pair is attached to the surface while the other vortex lifts off. The side on which the vortex is attached has a relatively lower pressure due to the suction created by the low pressure in the vortex core, resulting in a net side force.

Previous studies have shown that the nature of the asymmetry can be controlled using controlled surface imperfections (Taligoski et al. 2015) and can be significantly nullified using a highly polished surface (Mahadevan et al. 2018). In the case of significantly reduced surface imperfections (polished surface, Mahadevan et al. 2018) and absence of surface imperfections (numerical studies, Degani and Tobak 1992; Taligoski et al. 2015), the two vortices remain symmetric even at high angles of incidence. A numerical study by Guan et al. (2009) suggests that the ratio of imperfection height to the local boundary layer thickness is an important parameter in determining the initiation and growth of vortices. On the other hand, if the surface imperfection size exceeds the boundary layer significantly, such as for protrusions, it can considerably affect the aerodynamic loads of the vehicle. For instance, the aerodynamic load produce by surface protuberances has been a significant focus of study in launch vehicles such as Ares I, Ares V (from the US) and Epsilon (from Japan) (Harada et al. 2019).

The problem of vortex asymmetry at high angles of incidence is of significant practical interest and, therefore, has led to many experimental and numerical investigations. In spite of these efforts, many fundamental questions regarding the nature of instabilities causing vortex asymmetry remain unanswered. At the same time, a number of active and passive flow control techniques such as blunting, blowing, and a variety of boundary layer trips have been successfully implemented to reduce the magnitude of side-force (Kumar et al. 2005, 2008; Malcolm 1991). US Air Force Academy (Fagley and McLaughlin 2014; Porter et al. 2012) has recently developed a closed-loop flow control of a tangent ogive at high angles of incidence using plasma actuators. Their results suggest that with appropriate manipulation of asymmetric vortices, one can increase the maneuverability of the flight vehicle. Our recent study (Mahadevan et al. 2018) on the effect of controlled surface imperfections on the development of vortex asymmetry on a conical body has helped us

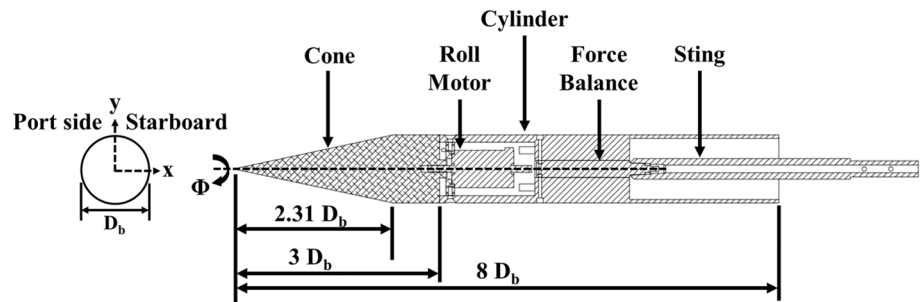
identify the length scales and most receptive locations on the forebody for maximum control effectiveness.

In the case of a long cone–cylinder body, the primary vortex pair at the forebody tip will eventually lift-off and be replaced by newly formed secondary shear-layer vortex pairs. Xueying and Yankui (2004) showed the development of multiple vortex pairs and their effects on the sectional side-force distribution. They correlated the vortex flow pattern to the sectional side-force distribution and divided the flow-field into different regimes. The inception regime is the forebody area where the sectional side-force increases and reaches a maximum. The triple vortex regime where the primary vortex on one side lifts off and formation of secondary shear-layer vortex begin. This results in a decrease in the side-force, which reaches a minimum. Eventually, the secondary shear-layer vortex also lifts off, and another new vortex starts forming from the shear-layer, which causes the sectional side-force coefficient to fluctuate. This sinusoidal pattern of increase and decrease in sectional side-force repeats, and the magnitude of the side-force eventually decays after three or four cycles.

The numerical study of Porter et al. (2012) shows that smaller coherent structures from the shear layer feed the primary vortex structure in contrast to the formation of secondary structures reported by Xueying and Yankui (2004). Ward and Katz (1989) used a laser-induced fluorescence technique to track the vortex core development along the length of the ogive body at 45° angle of incidence and different Reynolds numbers. From their study, it was evident that as the left-side primary vortex lifted off the right-side primary vortex started moving towards the center of the body. In some instances, they have shown that the primary vortex breaks apart and starts coalescing into distinct centers, and further downstream, they interlace and recombine to form a single structure.

It is evident from the ongoing discussion that the flow over a long slender body is very complex in nature and involve the evolution of multiple vortex pairs. The development of the flow-field can vary between studies due to the sensitivity to small scale surface imperfections. The goal of the present study is to experimentally investigate the flow-field over a generic slender body consisting of a conical fore-body and a cylindrical aftbody. The main objectives of the present experimental investigation are to understand the role of secondary shear-layer vortices in the development of primary vortices over the length of the body and the resultant side force. The flow over the entire body was measured and analyzed using Particle Image Velocimetry (PIV). The forces experienced by the body at different flow conditions were measured using a six-component strain gauge balance.

Fig. 2 Cross-sectional view of the cone–cylinder model



2 Experimental methodology

2.1 Test model

A long slender body of base diameter $D_b = 50.4$ mm and length of $8D_b$ was used in the current study. The cross-sectional view and the geometry specifications are shown in Fig. 2. The cone tip has a semi-apex angle of 12° and a small bluntness ratio (r_{tip}/D_b) of 0.01 to avoid a sharp tip. The test model was manufactured in two parts, with the assembly line located at $z/D_b = 3$. The first part consists of the conical forebody with part of the cylinder aftbody, and the second part consists of the remaining aftbody. The cone–cylinder junction was fabricated to have a smooth transition from cone to the cylinder with a nominal sharp edge without any prescribed radius. The first and second parts of the cone–cylinder model were carefully assembled with a very small intermediate gap (measured using a feeler gauge) of approximately 0.05 mm. The model was carefully fabricated with tight tolerances to minimize surface imperfections using Aluminum 6061. The test model was anodized black to reduce reflections while conducting PIV tests, which further reduced surface imperfections. The nose cone part of the model was attached to a stepper motor (step size = 0.225°) to achieve different roll orientations. The aftbody had the provision to hold both the stepper motor and the strain gauge balance. The strain gauge balance was attached to a support sting via an angle of incidence (high alpha) adapter. Different adapters helped in achieving different ranges of the angle of incidence for the same range of movement of the sting. The maximum test section blockage at any cross-section, with the cone–cylinder model, was 2.5%, at 50° angle of incidence.

2.2 Experimental facility

Experiments were conducted in the FSU Low-Speed Wind-Tunnel Facility. It is a conventional suction type open-loop wind tunnel driven by a 200 hp fan, which can reach up to 1700 rpm. The test section dimensions are (height) 762 mm $\times$ (width) 762 mm $\times$ (length) 1524 mm with acrylic

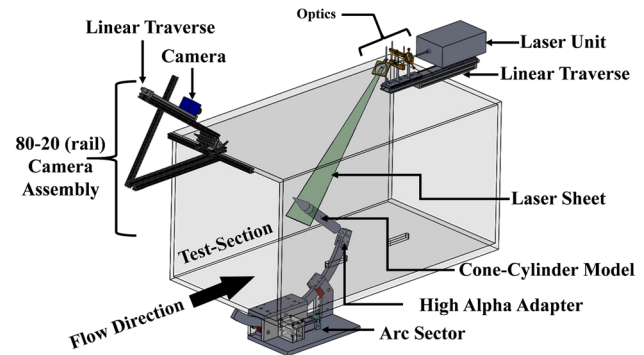


Fig. 3 Test model assembly in the FSU low speed wind tunnel test-section with PIV Setup

windows allowing optical access from all sides. Acrylic material light transmission is $> 92\%$, and the corresponding refractive index is 1.49. The inlet has a contraction ratio of 9:1 and is fitted with a series of flow conditioners resulting in a low turbulence intensity of less than 0.5 %. The freestream velocity can be accurately varied from 2 to 80 m/s and is regulated by changing the fan rpm using a variable frequency drive. The flow angularities in the pitch, $\Delta\alpha$, and in the yaw plane, $\Delta\beta$, of the test section are $\leq 0.2^\circ$. More details about the facility and its calibration are included in the reference (Guha 2017). The velocity is monitored using a pitot-static probe mounted at the entrance of the test-section, which accounts for the test-section model blockage. The model was attached to the high alpha adapter (0° – 20° and 20° – 50° adapters) in the arc sector assembly shown in Fig. 3. A digital inclinometer was placed on a flat surface at the top of the high alpha adapter to measure the angle of incidence. The flat surface was parallel to the axis of the body. The arc sector assembly used for pitch control was placed underneath the test section and emerged through a slot in the test section floor. At an angle of incidence of 50° , the model nose was at least 10-inches away from the tunnel wall, and hence wall corrections were not necessary. For the current experimental study, the freestream velocity was kept constant at 40 m/s, and the corresponding Reynolds number (based on D_b) is 1.3×10^5 .

2.3 Force and moment measurements

A six-component internal strain gauge balance (Aerophysics Research Instruments – 0.5 MKXV Internal Balance) was used to measure the forces and moments. The maximum force ratings are 444 N, 222 N, and 222 N for normal, axial, and side-forces, respectively. The maximum moment ratings are 24 N-m, 10 N-m, and 6 N-m for pitching, yaw, and roll-moments, respectively. The strain gauges used were of self temperature compensation type to ensure that the force balance was in thermal equilibrium over the course of testing. To acquire the data, a National Instruments data acquisition card (NI PCIe-6320) interfaced with an SCXI-1000 chassis, an SCXI-1530 strain gauge module, and an SCXI-1314 terminal block (to which the force balance was connected) were utilized. The side force “S+” bridge was oriented to face the left (/port) side of the body, and the normal force “N+” bridge was oriented towards the positive y -direction. An in-house calibration was performed by loading (and unloading) weights on the individual bridges and recording the corresponding voltage outputs. A least-square global regression method was used to generate a 96×6 AIAA standard calibration matrix. An in situ check load test was conducted before the experiments. Sting-balance deflections were measured using known loads to apply for pitch correction. For the measured loads, the deflection constants were very small, so no correction was needed. In the present study, the focus was on side force measurements and less on drag. Therefore, base pressures were not measured to apply for a base drag correction. For all cases, data were acquired for 30 seconds for a sampling frequency of 100 Hz to obtain a statistically accurate ensemble average.

2.4 Planar particle image velocimetry

Planar particle image velocimetry was used to obtain flow-field information for three roll orientations of 0° , 45° , and 180° , at 40° angle of incidence. The flow was seeded using glycol particles having dimensions on the order of $2\text{--}5\ \mu\text{m}$. These particles were illuminated using a Quantum[®] Nd:YAG double pulsed laser. The incident ray was focused using a plano-convex spherical lens (focal length of 1000 mm) and fanned into a sheet using a plano-concave cylindrical lens (focal length of 25 mm). The sheet was reflected into the test-section using a laser-line mirror. The thickness of the laser-sheet in the window of interrogation was approximately 2 mm. Images were acquired using a 5.5-megapixel sCMOS camera having an sCMOS array of 2560×2160 pixels. A 105 mm Nikon lens was used for the cylindrical part to cover a larger field of view since the asymmetric vortex

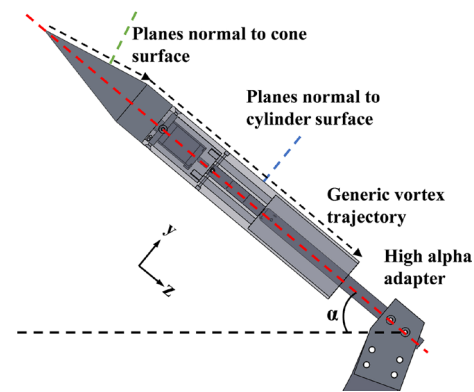


Fig. 4 PIV planes orientation with respect to the model

tends to lift off from the body. A 180 mm Nikon lens was used for the conical part to focus on the initial small vortices.

In the present study, two individual camera calibrations (one for the cone portion and the other for cylinder portion) were conducted. The laser, optics, and camera were mounted separately on motorized traverse systems, Fig. 3. The linear traverses were connected to stepper motors having a step resolution of 0.1 mm. The camera—linear traverse rested on a separate rig with the flexibility to change the angular orientation of the camera axis. The axis of the camera (looking downstream) was set parallel to the model for planes on the cylinder. It was set parallel to the cone-surface for planes on the cone. Correspondingly, the laser sheet was normal to the axis of the body for cylinder-planes and normal to the cone-surface for cone-planes (Fig. 4), to capture the vortex traveling along the contour of the surface. The traverse system holding the laser and optics was placed on top of the diffuser-section. The two traverses were moved in a coordinated manner so that the camera retained its focus on the field of view at different axial locations, eliminating the need for repeated calibrations.

The DaVis 8.4.0 software (LaVision Inc.) was used for data acquisition and processing. The time interval between the double pulses was set to $16\ \mu\text{s}$ to acquire a mean displacement of ≥ 6 pixels. Statistically ensemble data were acquired at a rate of 15 Hz. Vector calculations were done using a multi-pass, decreasing correlation window size algorithm. The initial passes had a correlation window size of 64×64 pixels with a 50% window overlap. The final passes had a window of 32×32 pixels with a 75% overlap. The resulting vector resolution for the cone part was 0.3 mm ($0.006 D_b$), and for the cylinder part was 0.6 mm ($0.012 D_b$), along both the axes.

Data were acquired at various cross-stream planes (z -locations) along the length of the body. The normalized distance ($\Delta z/D_b$) between consecutive planes for the cone part was

0.03, and the cylinder part was 0.06. Above-mentioned PIV setup would be termed as "PIV type-1" for the rest of this study. To obtain spatially denser information about the flow-field, "PIV type-2" was performed for 180° roll cases at 40° angle of incidence. The experimental setup is the same for the two types, with the difference being the distance between the consecutive planes and the number of images acquired. For the PIV type-2 cases, the distance between consecutive planes ($\Delta z/D_b$) for the cone part and the cylinder part was 0.01. A 1000 instantaneous snapshots were acquired for cases corresponding to PIV type-1. These results showed that 300 snapshots were required for the convergence of the mean flow-field. Therefore, 300 images were acquired for cases corresponding to PIV type-2 to optimize test time.

2.5 Measurement uncertainty

The inclinometer used to measure the angle of incidence has an error of $\pm 0.05^\circ$ for the range of 0° to 10° , and for the angle of incidence greater than 10° , the error is $\pm 0.1^\circ$. The dynamic pressure was recorded using a SETRA Manometer, which has an uncertainty of $\pm 0.1\%$ of full scale. The uncertainty in the tunnel velocity was determined to be ± 1 m/s. The strain gauge balance has a maximum error of $\pm 0.02\%$ of full scale for all elements. In the present study, uncertainty in side-force was calculated by the method suggested by Mahadevan et al. (2018), and the measurement uncertainty for the side-force coefficient was found to be $\pm 0.011 C_s$.

The uncertainty in the PIV flow-field is acquired from the DaVis software, which utilizes the correlation statistics method suggested by Wieneke (2015). Uncertainty in velocity components is less than 1% except in the vicinity of regions where the velocity magnitude is nearly zero. This stems from the difficulty faced by the PIV algorithm to resolve very small velocity components. Similar characteristics are observed for all cases of PIV type-1 and type-2.

3 Results and discussion

3.1 Side force characteristics

Side force characteristics are presented in the form of side force coefficient (C_s), which is obtained using the following equation:

$$C_s = \frac{SF}{0.5 \times \rho_\infty \times ((U_\infty)^2) \times (\pi/4) D_b^2} \quad (1)$$

where SF is the measured side-force, ρ_∞ is the freestream density, and D_b is the base diameter of the cone-cylinder. The variation in the side-force coefficient with the variation in the angle of incidence is shown in Fig. 5a. The side-force coefficient is zero at small angles of incidence where the flow is either attached or separates symmetrically. The side-force starts developing at $\alpha = 24^\circ$ and thereafter increases with the rise in the angle of incidence, reaching a peak value of $C_s = 5.4$ at $\alpha = 47^\circ$. This indicates the separation has become asymmetric, where one vortex is closer and the other away from the body. After reaching the peak, a sharp drop is observed in the side-force coefficient indicating the onset of vortex shedding. The onset of vortex asymmetry occurs at an angle equal to the cone apex angle, and the results are very similar to those seen in previous studies on conical fore-bodies (Taligoski et al. 2015, Mahadevan et al. 2018). Experimental studies reviewed by Hunt (1982) state that slender bodies with $L/D \geq 10$ are necessary to alter the onset of asymmetry significantly.

In the current study, roll-search was conducted to identify the roll orientations corresponding to a maximum and a minimum side-force magnitude at a given angle of incidence. For angles of incidence less than 24° , data were acquired for every 20° roll step, and at higher angles of incidence for every 9° roll step. The variation in side-force coefficient with roll-orientation for angles of incidence of 10° and 40°

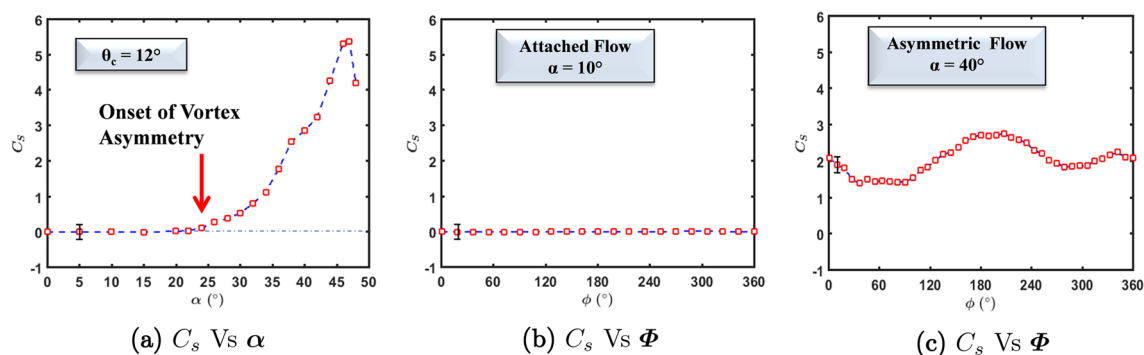


Fig. 5 Variation of side-force coefficient with angle of incidence and roll orientation

are shown in in Fig. 5b, c, respectively. At $\alpha = 10^\circ$, the side-force coefficient is zero at all roll orientations. At $\alpha = 40^\circ$, the side force varies periodically with a maxima at $\Phi = 180^\circ$ and a minima at around $\Phi = 45^\circ$. An interesting observation is that the side-force always remains positive, unlike a sinusoidal behavior observed on conical fore-bodies (Taligowski et al. 2015; Mahadevan et al. 2018). In the present study, only the front part of the body consisting of the entire cone, and a fraction of the cylinder was rolled, while the aftbody was kept in the same position. Since previous studies have shown that surface imperfections at the cone tip, which are of the order of the local cross-stream boundary layer, are the main source of vortex asymmetry. The present results suggest that the entire body may play a role in the total flow asymmetry over the body and not just the cone tip. Hence, a periodic behavior is observed since a fraction of the body was rotated, but a change of sign is not observed as the roll orientation of the major section of the cylinder remained constant. The PIV flow-field (to be discussed) sheds more light on this observation and argument.

3.2 Characteristics of flow-field over a slender body

3.2.1 Generic evolution of vortices

The PIV time-averaged results for selected planes on the slender body model, at $\alpha = 40^\circ$ and $\Phi = 180^\circ$, are shown in Fig. 6. These results correspond to the PIV Type-1 cases. The first three locations (Fig. 6a–c) were acquired on the conical forebody and the remaining three (Fig. 6d–f) were acquired on the cylindrical aftbody. The results show the generic characteristics of the flow-field. The results are presented in the form of non-dimensional streamwise vorticity fields. The expressions for calculating dimensional and non-dimensional streamwise vorticity (ω_z and ω_z^*) are shown in Eqs. 2, 3, respectively.

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (2)$$

$$\omega_z^* = \frac{\omega_z D_b}{U_\infty} \quad (3)$$

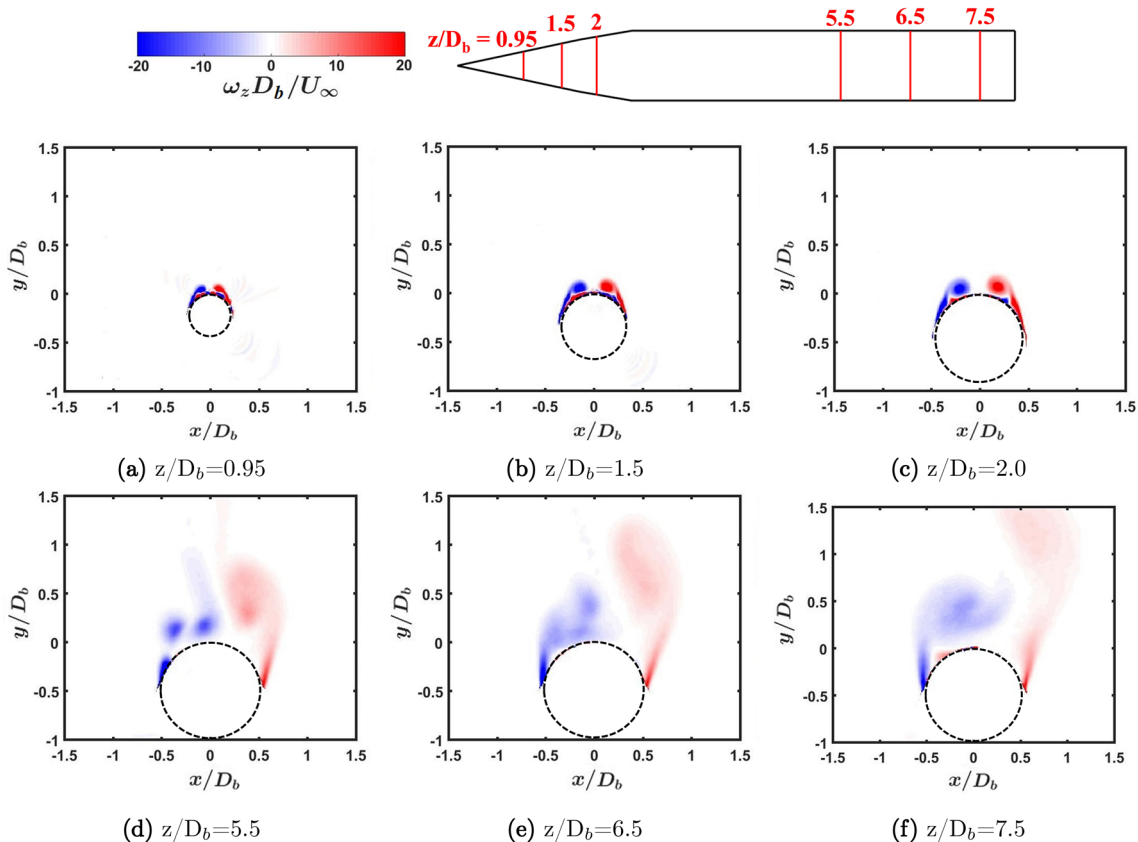
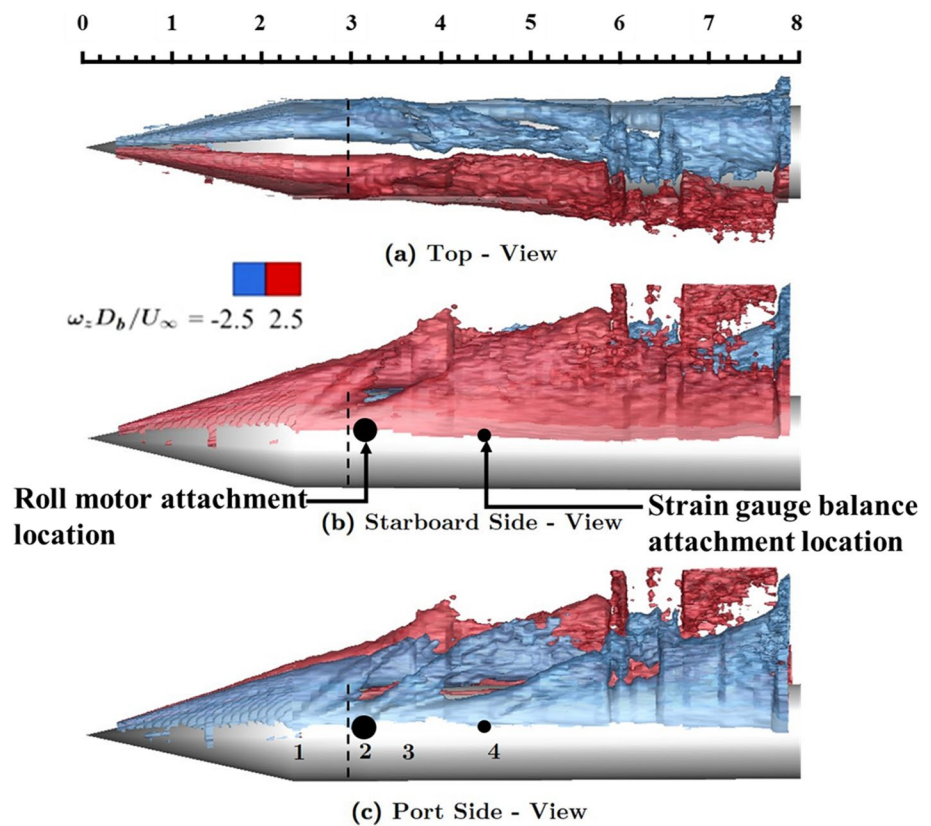


Fig. 6 Non-dimensional mean vorticity fields at selected locations on the model at $\alpha = 40^\circ$ and $\Phi = 180^\circ$

Fig. 7 Non-dimensional mean z -vorticity iso-surfaces at $\alpha = 40^\circ$ and $\Phi = 180^\circ$



A pair of small counter-rotating vortices develop at the cone-tip and gradually grow in size over the length of the body. The vortices are nearly symmetric on the cone but develop significant asymmetry, while over the cylinder. At $z/D_b = 7.5$, the port side vortex (with negative vorticity) is relatively closer to the surface and seems to encompass the entire width of the body. Whereas, the starboard side vortex (with positive vorticity) has moved far away from the body in both horizontal and vertical directions. At the location $z/D_b = 5.5$, a secondary shear-layer vortex develops from the shear layer and merges with the primary vortex. The resultant vortex after merging is observed to be highly asymmetric. To understand these various flow features, we first look at the vorticity iso-surface over the entire body, generated from the PIV Type-2 results. It provides an overall picture of the flow-field, followed by discussion on the characteristics of the vortex themselves and their variation with different roll orientation.

3.2.2 Vorticity iso-surfaces

Vorticity iso-surfaces are created from the assembly of a total of 80 planes of the flow-fields obtained from PIV Type-2. Top, starboard side and port side views of the iso-surfaces for $\Phi = 180^\circ$ are shown in Fig. 7. Vorticity iso-surfaces help in identifying the initiation (from nose tip), development,

and the breakdown of the primary vortices over the entire cone-cylinder model. As discussed in the earlier section, the nearly symmetric primary vortices on the nose-cone become highly asymmetric as they propagate over the cylindrical section. Also, at multiple locations along the cylindrical aftbody, secondary vortices generated from the shear layer merge with the primary vortex. Finally, towards the aftbody end, the vortices start to dissipate rapidly, and the breakdown process is initiated.

It is observed that the first set of secondary vortices develop right at the cone-cylinder junction. A vortex develops on both starboard and port side, and are non-identical. It is possible that due to a sharp change in geometry, the flow locally separates, leading to the formation of secondary vortices. In such a scenario, it will be interesting to see if those secondary vortices are eliminated with a smooth change in geometry, as in an ogive-cylinder body. There can be additional sources of this asymmetry, like freestream turbulence (incoming flow asymmetry), asymmetric flow development over the cone, or a combination of both.

The second set of secondary shear-layer vortices, seem to develop on both sides, from the assembly line of the two halves of the model. As mentioned in the test model description, the intermediate spacing at this joint was approximately 0.05 mm. Closely following the assembly line was a line of tapered holes, meant for holding the stepper motor, that

rolled the front part of the model. These holes were filled with putty and cleaned to have a smooth surface. In spite of all the care taken, it is possible to incur minute surface imperfections leading to local flow distortions. The location of the junction and the tapered holes are demarcated in the figure. The vortex developing on the port and starboard sides are asymmetric. The asymmetry could have stemmed purely from the incoming flow asymmetry or the surface asymmetry at the hole locations, or both.

The third set of secondary vortices develop slightly downstream of the second set. While it is visible on the port side, it is not visible on the starboard side. The second set of threaded holes were located at the point where the integral force balance was attached to the model, demarcated in the figure. A fourth vortex is observed at this location on the port-side, Fig. 7c. It seems to merge with the third vortex on the same side. The development of multiple vortices corroborates with the computational study by Xueying and Yankui (2004), which states that these vortices significantly affect the sectional side-force distribution. It is clear from the results that a sudden change in geometry like at the cone–cylinder junction will lead to the development of such vortices. However, it is not completely clear if the rest of the pairs are characteristics of the natural flow or due to surface imperfections. Nevertheless, the observations tend to lean towards the latter. A similar study by Harada et al. (2019) has also shown that surface defects (such as protrusion) generate a secondary shear-layer vortex, which merges with the primary vortex. This vortex merging process induces a surface pressure asymmetry which leads to a side force. Therefore, the location and the cause of initiation of secondary shear-layer vortices could be different for different applications. Still, the process of merging and its effect on aerodynamic forces will be very similar. For most experimental investigations and real applications, such joints and local surface imperfections will always be present in one or another form.

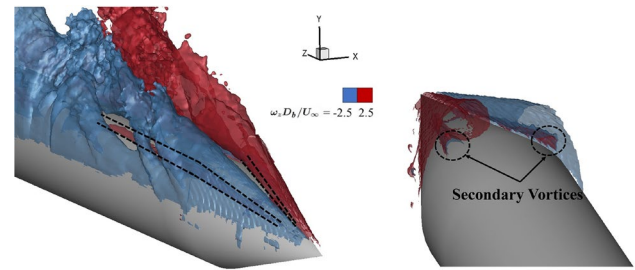


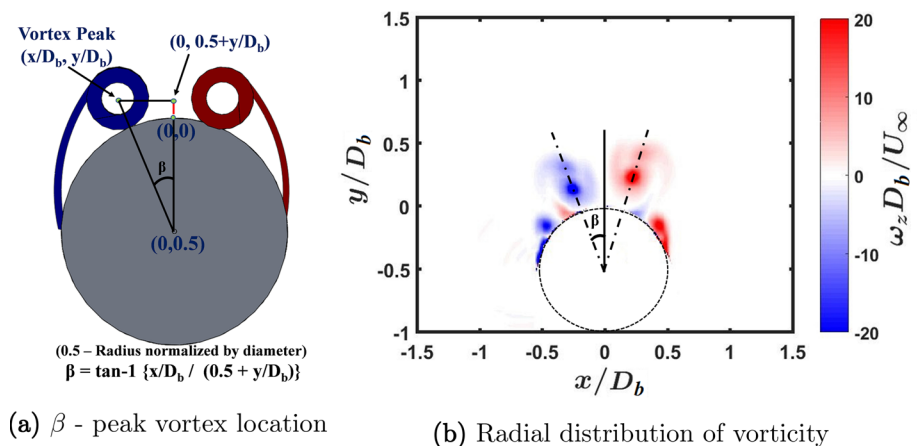
Fig. 8 Non-dimensional mean z -vorticity iso-surfaces over the cone at $\alpha = 40^\circ$ and $\Phi = 180^\circ$

An isometric view of the iso-surface with higher transparency and a rear view of the cone-flow is shown in Fig. 8. secondary vortex pair is found to develop between the primary vortex pair and the surface. The trajectory of the secondary vortex pair is shown by dotted lines in the isometric view and dotted circles in the rearview. This vortex pair is counter-rotating with respect to the primary vortex pair above it. These line vortices seem to extend over the entire cone and certain sections of the cylinder. The characteristics of these trapped vortices are probably dictated by the primary vortices. Since they are closest to the surface, they will have an important impact on the side-force development. The following discussion looks at the characteristics of the vortices themselves.

3.2.3 Radial distribution of vorticity

For selected roll orientations, the radial distribution of vorticity along the vortex core and the corresponding location (r/D_b , β) of the vorticity peak is determined (Fig. 9). These results correspond to PIV Type-1. The radial location (r/D_b) of the peak vorticity indicates how close the vortex position is with respect to the surface of the body. The circumferential location (β) indicates the angle

Fig. 9 Estimation of peak vortex circumferential location and radial distribution of vorticity ($z/D_b = 3.5$)



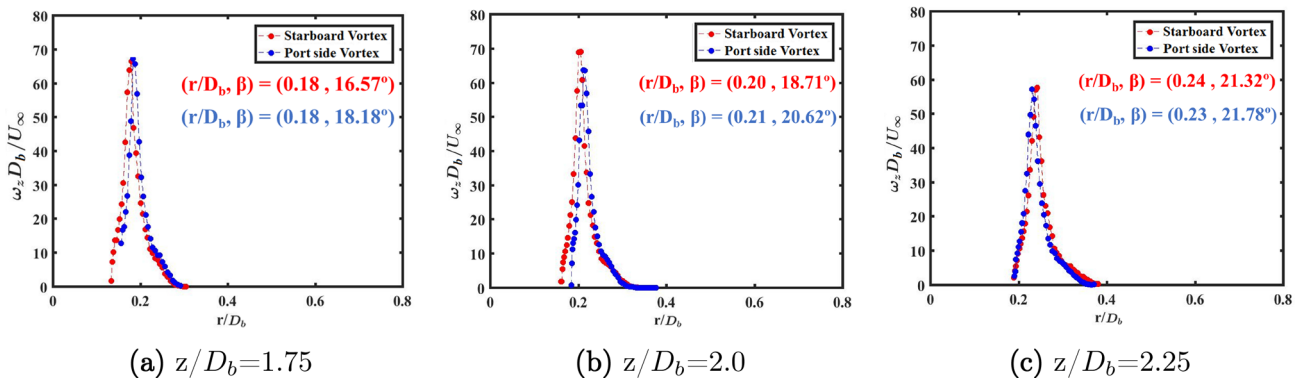


Fig. 10 Streamwise vorticity distribution at selected locations on the cone for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

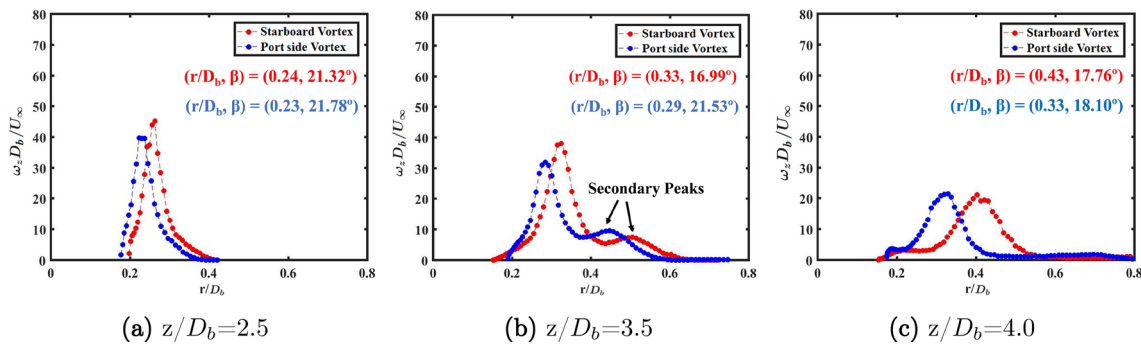


Fig. 11 Streamwise vorticity distribution at selected locations on the cylinder for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

between the vertical axis and peak vortex location. In Fig. 10, z/D_b of 1.75, 2 and 2.25 represents the cone part. The radial locations (r/D_b) of the right and left vorticity peaks are nearly the same.

On the cone ($z/D_b = 1.75, 2, 2.25$), the circumferential locations of the starboard and port vorticity peaks are slightly different though their radial locations are the same. The starboard vortex " β -location" is positioned closer to the vertical axis, and the port vortex is positioned farther away from the vertical axis. One plausible reason for such a dissimilar circumferential location of starboard and port side vortices on the cone is the circumferential location of the flow separation. It may be possible that the flow separation on the port side is occurring earlier in comparison to the starboard side. Selected plots correspond to vorticity distribution on the cylindrical region (z/D_b of 2.5, 3.5, and 4) are shown in Fig. 11. At $z/D_b = 3.5$, secondary peaks are visible on both starboard and port-side vorticity distributions. A shift in peak vorticity value towards the right in the plots suggests the peak vortex location is moving away from the surface. The strength of the secondary peaks is lower than the primary peaks. These secondary peaks correspond to the secondary shear-layer vortices, which form on both sides of the cylinder and merge with

their respective primary vortices. At $z/D_b = 4$, the peak vorticity is almost the same for the port side and starboard vortex, whereas, their radial and circumferential locations are significantly different.

3.2.4 Variation of vortex size and location with roll orientation

In the previous discussion, the mean vortex characteristics were presented using the 180° roll-orientation case that produced the highest side-force coefficient. In this section, the variations in the mean characteristics with the roll-orientation of the forebody are presented. Figure 12a shows the non-dimensional vorticity fields at a cross-section near the end of the cone (at $z/D_b = 2$) for $\Phi = 0^\circ, 45^\circ$, and 180° . Figure 12b shows the corresponding fields at a cross-section near the end of the cylinder (at $z/D_b = 7.5$). Figure 12a shows that the vortices are nearly symmetric and the development is almost identical on the conical part for all three roll cases. Figure 12b shows that the development on the cylindrical part has differences, especially between $\Phi = 180^\circ$ and the rest. The extracted vorticity distributions in Fig. 13, show that the width of the vortex and the radial location of the vorticity peak are nearly invariant with Φ on the cone. Only

Fig. 12 Comparison of vortex development for $\alpha = 40^\circ$ and $\Phi = 0^\circ, 45^\circ$, & 180° roll orientations

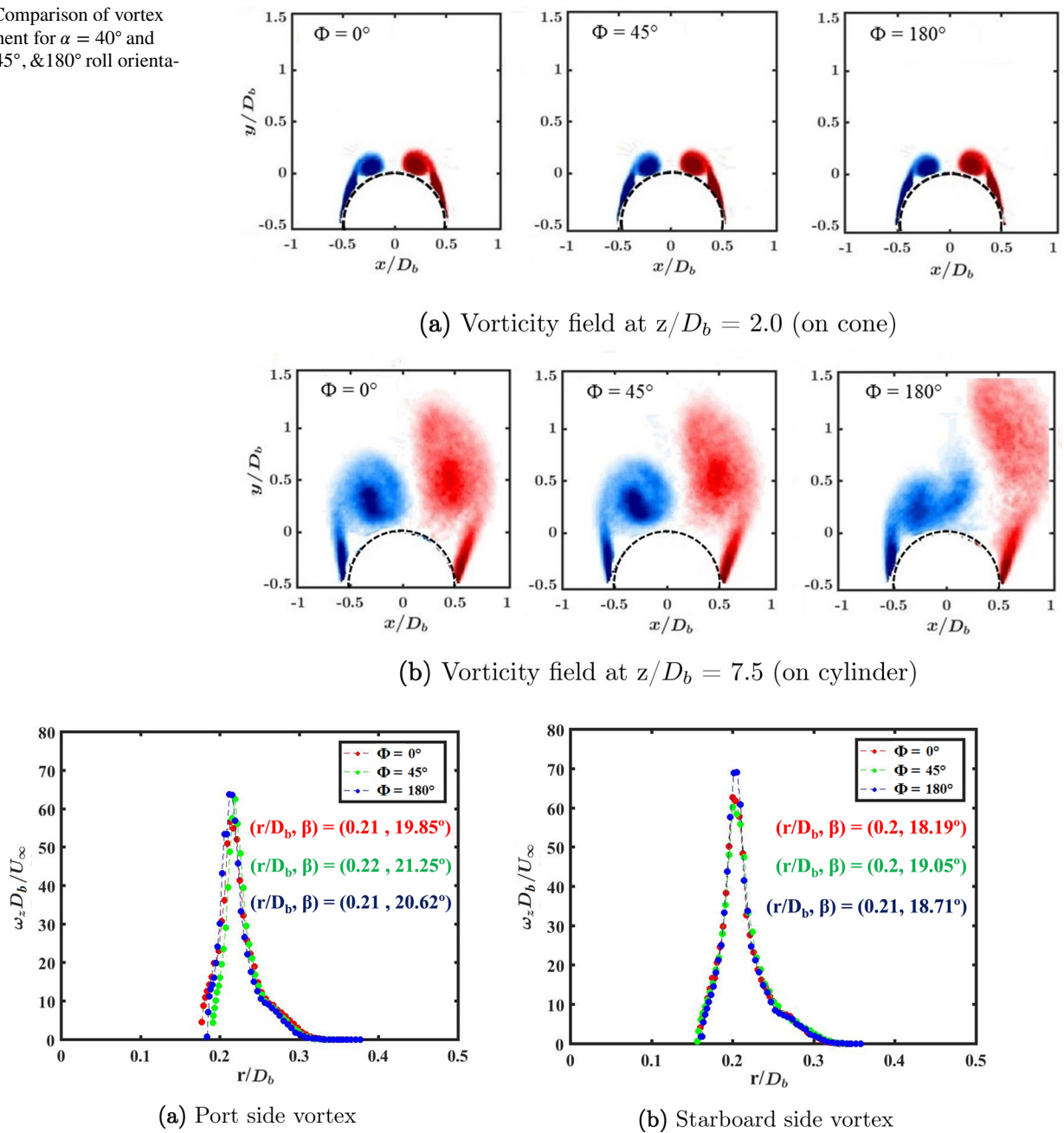


Fig. 13 Comparison of vorticity distribution at $z/D_b = 2.0$ (on cone) for $\alpha = 40^\circ$ and $\Phi = 0^\circ, 45^\circ$, & 180° roll orientations

minor variations are observed in the circumferential location (β) and peak vorticity values. Whereas on the cylinder, Fig. 14, the peak's radial, and circumferential location vary significantly with Φ . The width of the vortex is very similar for 0° and 45° roll orientation both on the starboard and the port-sides. However, the starboard vortex for the $\Phi = 180^\circ$ is larger, diffused and farther away from the body in comparison to $\Phi = 0^\circ$ and 45° . The port side vortex for $\Phi = 180^\circ$ is relatively closer to the body and much shorter in width, in comparison to the starboard. Visually from Fig. 12b, it

seems that the vortex merging process has not reached completion for $\Phi = 180^\circ$, and the vorticity distribution is comparatively less compact.

3.2.5 Variation of peak vorticity with roll orientation

The highest non-dimensional mean vorticity values on the port and starboard sides at different cross-stream planes for the three roll cases are shown in Fig. 15. The blue symbol

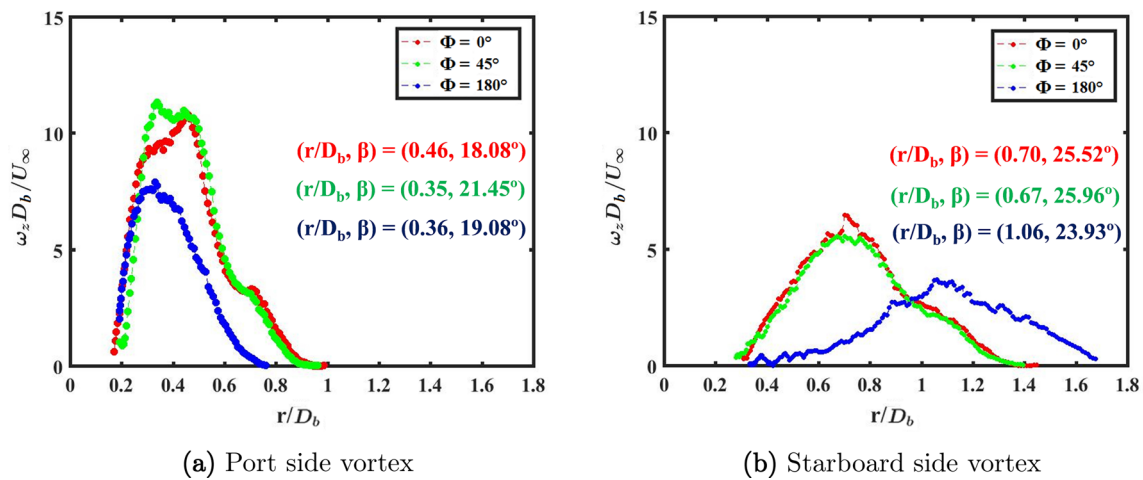


Fig. 14 Comparison of vorticity distribution at $z/D_b = 7.5$ (on cylinder) for $\alpha = 40^\circ$ and $\Phi = 0^\circ, 45^\circ$, & 180° roll orientations

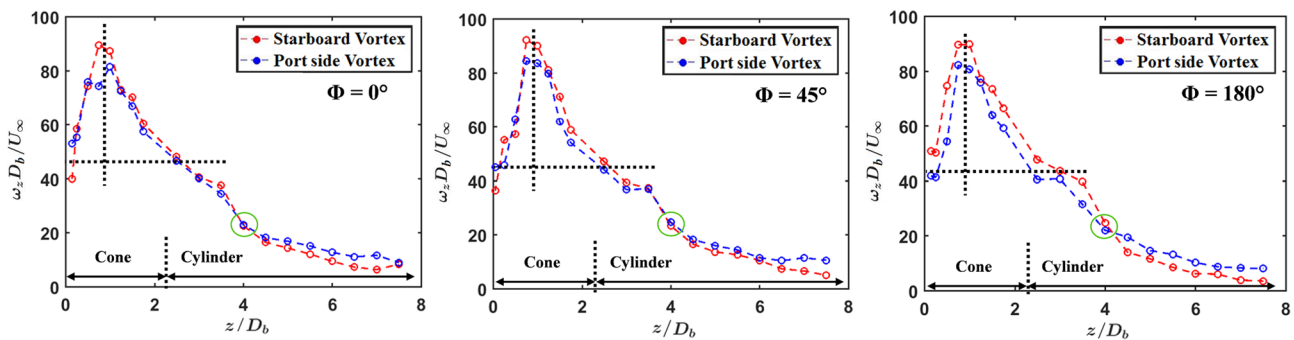


Fig. 15 Peak non-dimensional vorticity at different measurement planes for $\alpha = 40^\circ$ and $\Phi = 0^\circ, 45^\circ$, & 180° roll orientations

indicates the peak vorticity on the port side, and the red symbol denotes the peak vorticity on the starboard side. For all the cases, vortices of significant strength develop at the first measurement plane ($z/D_b = 0.3$) near the cone tip. At all the three roll orientations, the peak vorticity increases till $z/D_b = 1$, reaches maxima, and decreases thereafter. The attainment of vorticity peak at $z/D_b = 1$, and general variation in vorticity till the cone–cylinder junction, is similar for all three roll orientations.

The starboard side has a slightly higher maximum value than the port side for all the cases. A consistent difference in the peak values along the entire length of the body is observed at $\Phi = 180^\circ$, which showed the highest side-force. For all three roll orientations, the peak vorticity of the port-side vortex gains strength and eventually equals the peak vorticity of the starboard-side vortex at $z/D_b = 4$ (marked with a green circle in Fig. 15), beyond this point the port-side vortex becomes stronger than the starboard-side vortex. For the $\Phi = 0^\circ$ case, the peak vorticity values on the port and starboard-sides again become equal towards the end of the model. Whereas, for the $\Phi = 45^\circ$ and $\Phi = 180^\circ$, the peak

vorticity values remain different. This gain in peak vorticity on the port side could be due to the merging of the secondary shear-layer vortex and the primary vortex. These results are very similar to those observed on a conventional cone body (Mahadevan et al. 2018 and Taligowski et al. 2014). They showed that vortices initiate near the cone tip, the peak vorticity values increase, attain maxima, and then decrease on the conical section itself.

3.2.6 Variation of circulation with roll orientation

To assess the cumulative vortex strength, circulation was calculated using the equations shown below,

$$\Gamma = \oint \omega_z dA \quad (4)$$

$$\Gamma = \sum_{i=1}^N \omega_{z,i} A_i \quad (5)$$

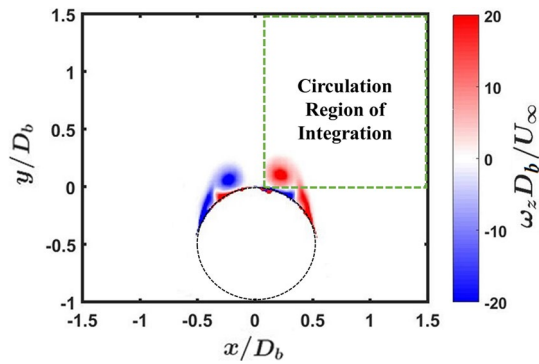


Fig. 16 Region of interrogation for calculating circulation ($z/D_b = 2.0$)

where the area A_i is the square of the final vector resolution. Figure 16 shows the region of integration considered for determining the circulation. Figure 17 shows the non-dimensionalized circulation values for the three roll orientations at different measurement planes. The blue and red symbols represent the port-side and starboard-side vortices, respectively. Data corresponding to the last two planes for $\Phi = 180^\circ$ are not included, since a portion of the starboard-side vortex went beyond the PIV window of interrogation. For all three cases, circulation value increases along the length of the body. For $\Phi = 0^\circ$ and 45° , the circulation values are very

similar for the port and starboard-sides until $z/D_b = 4$. Values start to vary beyond it; however, they become equal again towards the end of the body. For $\Phi = 180^\circ$ case, variations begin early around $z/D_b = 2$ and a considerable difference in the circulation values are observed after $z/D_b = 4$. Data are absent in the final planes to show if the values become exactly equal, like 0° and 45° cases.

The variations in the port and starboard side circulation values may be due to the merging of the secondary vortices with the primary vortex pair. There are three locations at which the secondary shear-layer vortex merges with the primary vortex. The first merging process is complete around $z/D_b = 3$. The second and third merging process ends around $z/D_b = 4.5$ and 7. Around the location of the merging of the second secondary shear-layer vortex, a large difference in the circulation values seem to develop for all the three cases, with the largest being for $\Phi = 180^\circ$. The difference in the strength of the secondary shear-layer vortex developing from the two sides could have led to this effect. Our recent study on a conical fore-body (Mahadevan et al. 2018) also showed that a decrease in net circulation of the primary vortex on one side (port or starboard) leads to a proportional increase in the circulation of the other.

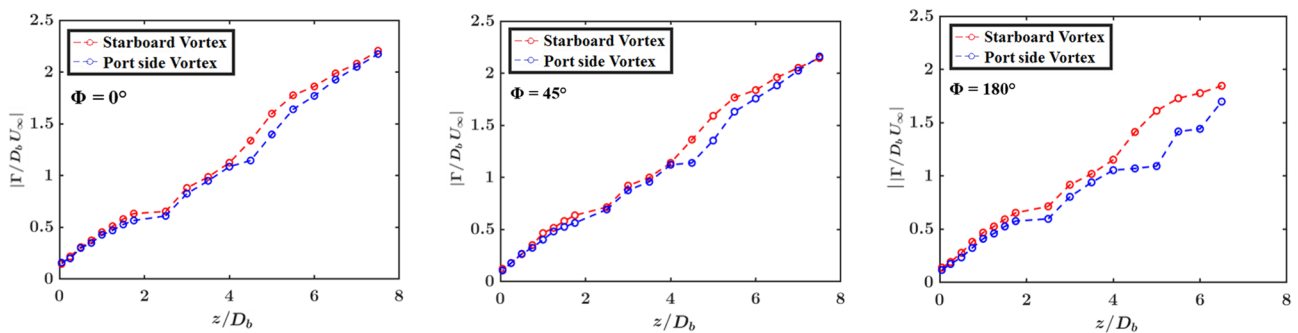


Fig. 17 Circulation at different measurement planes for $\alpha = 40^\circ$ and $\Phi = 0^\circ, 45^\circ, \& 180^\circ$ roll orientations

Fig. 18 Determination of the vortex size in the measurement plane ($z/D_b = 2.0$)

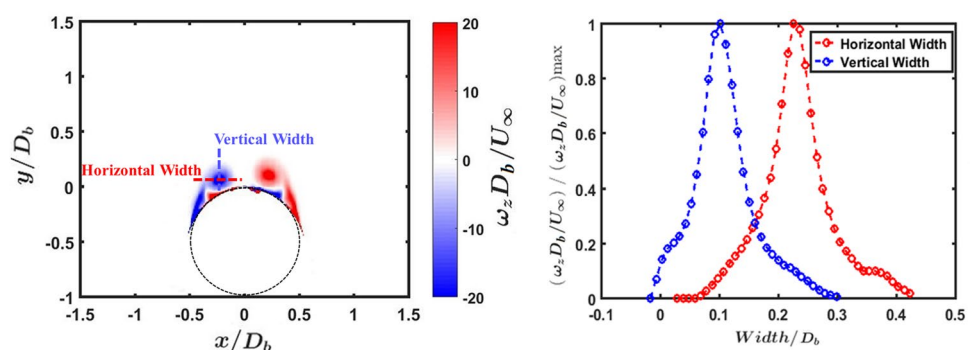
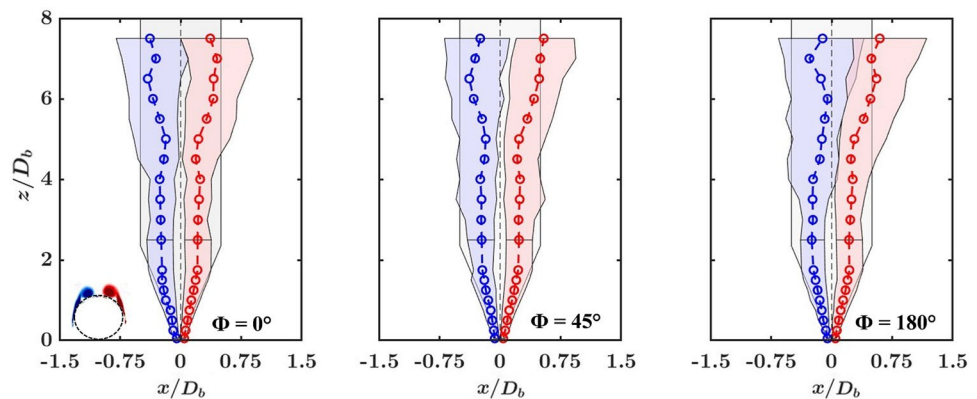


Fig. 19 Trajectory of peak vorticity and approximate boundary of the vortex core—along the flow direction



3.2.7 Trajectory of the vortices

Figure 18 shows the measurement of the size of the vortices. To obtain these dimensions at a particular plane (for port or star-board vortex), the vorticity values are extracted along the vortex core in the x/D_b and y/D_b directions, about the vorticity centroid. The resultant vorticity profiles are normalized using the peak vorticity value. The height and width are defined as the intermediate distance between the points where the vorticity value is 10% of the peak, in the respective profiles. The calculation of horizontal and vertical width at the different planes of measurement also indicates the variation in the shape of the vortex during its development. For example, at $z/D_b = 2$, the values of horizontal and vertical width are nearly the same indicating a circular

vortex at that plane as compared to an elliptical vortex at the downstream planes of measurement.

The trajectories of the vortices along the streamwise and normal direction of the flow (along z/D_b and y/D_b), for the three roll-cases, are shown in Figs. 19, 20, respectively. A trajectory is obtained by linearly interpolating between the positions of the mean peak vorticity value at different cross-planes. The blue and red dotted lines represent the port and starboard side's primary vortex trajectory, respectively. The area occupied by the vortex at different planes is shown by extracting the width (along x/D_b) and height (along y/D_b) of the vortices at these planes and overlaying them on the trajectory. Linear interpolation is made between the approximate width and height of the vortices at different planes. The blue and red regions correspond to the port-side and starboard-side vortices, respectively. The region occupied by the body is shown in gray to help in visualization. The area occupied by the vortices gradually grows in size along the length of the body. This is the result of gradual diffusion of the vorticity peaks and continuous assimilation of the separated shear layer.

For the roll cases of $\Phi = 0^\circ$ and $\Phi = 45^\circ$, the vortices are nearly symmetric with the central axis of the body, for the majority of its length. A small asymmetry is observed only near the trailing edge ($z/D_b \geq 6.5$). However, for the $\Phi = 180^\circ$ case, the asymmetry starts early at $z/D_b = 4$, and the vortices begin to deviate towards the starboard side. At the last plane, the port-side vortex occupies most of the region over the cylinder, whereas the starboard-side vortex has almost entirely moved off-axis. The vertical position of the vortex core does not change until z/D_b of 5.5 for all three roll cases, Fig. 20. The vortex cores (both port and starboard) for the $\Phi = 0^\circ$ case are attached to the surface, whereas for the $\Phi = 45^\circ$ case, the starboard vortex lifts from the body at z/D_b of 6. For the 180° case, the starboard vortex has entirely lifted off the surface at z/D_b of 6. It is evident from Fig. 20 that the width of the vortex has increased to almost

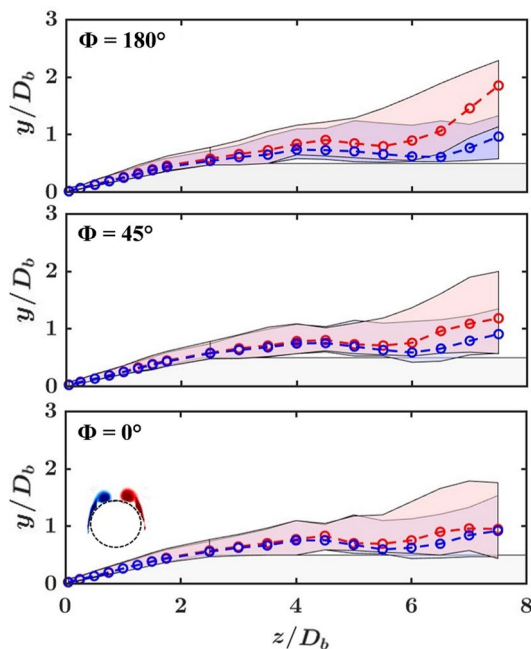


Fig. 20 Trajectory of peak vorticity and approximate boundary of the vortex core—normal to the flow direction

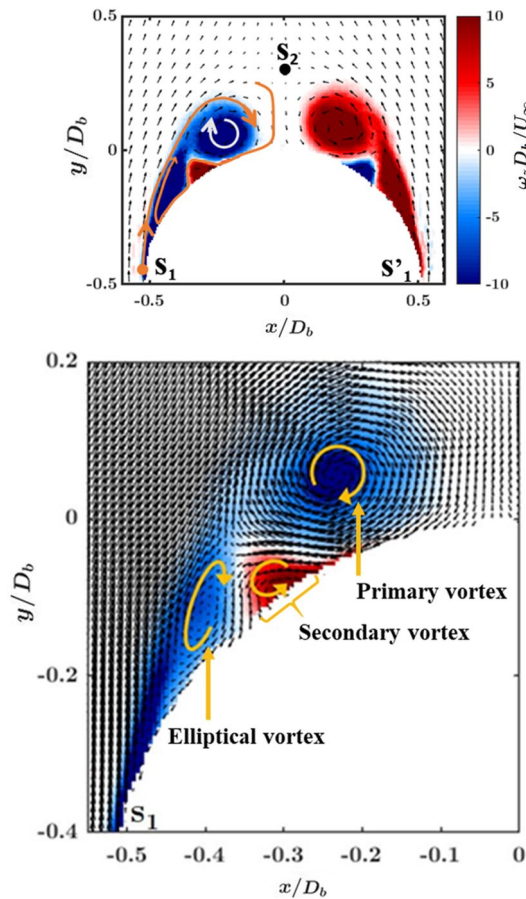


Fig. 21 Characteristics of symmetric flow over the slender body at $z/D_b = 2.31$ for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

$1.5D_b$ by the time it reaches to z/D_b of 7.5. For typical finned projectiles, the control surfaces are of the order of $0.8D_b$ to $1D_b$. In these situations, the control surfaces are likely to be engulfed by the vortices, thereby altering their effectiveness.

3.3 Near surface velocity and vorticity distributions

The characteristics of the flow close to the surface are discussed in this section. Three types of flows are observed on the cone–cylinder model. The first type is the nearly symmetric flow-field. The second type is the highly asymmetric flow-field, with one vortex dominating over the body. In the third type, secondary shear-layer vortices develop from the side of the body and merge with the existing primary vortices. The various characteristics of these three flow types are discussed in the following sub-sections.

3.3.1 Nearly symmetric vorticity field

Figure 21 shows the nearly symmetric vorticity field at the end of the cone and a zoomed view of its port-side flow-field. The cross-stream flow separates at S_1 on the port side (S'_1 on starboard) and forms the primary vortex. The saddle point between the two primary vortices is located at S_2 . The flow topology observed here is well studied and typically observed on slender bodies and delta wings at high angles of incidence (Sohail et al. 2011 and Hummel 1979). The primary vortex, secondary vortex, and a slightly elliptical vortex (emanating from the separated shear layer) are pointed out in Fig 21. The side-forces developing on the body will depend on the velocity distribution near the surface. It was challenging to extract the PIV data in the vicinity of the surface due to surface reflections. However, an attempt has been made to understand the near-surface velocity distributions by extracting profiles very close to the surface from the acquired PIV data.

The contour of data extraction is shown in Fig. 22. The extracted non-dimensional mean velocity (U_x and U_y) and vorticity (ω_z) profiles are shown in Fig. 23a, b, respectively. The value of U_x is almost zero near the stagnation point on the surface, directly below the saddle point S_2 (indicated by the point A_u). This point onward its magnitude, $|U_x|$, increases along the surface reaching a maxima $\approx |0.8|$ (B_u

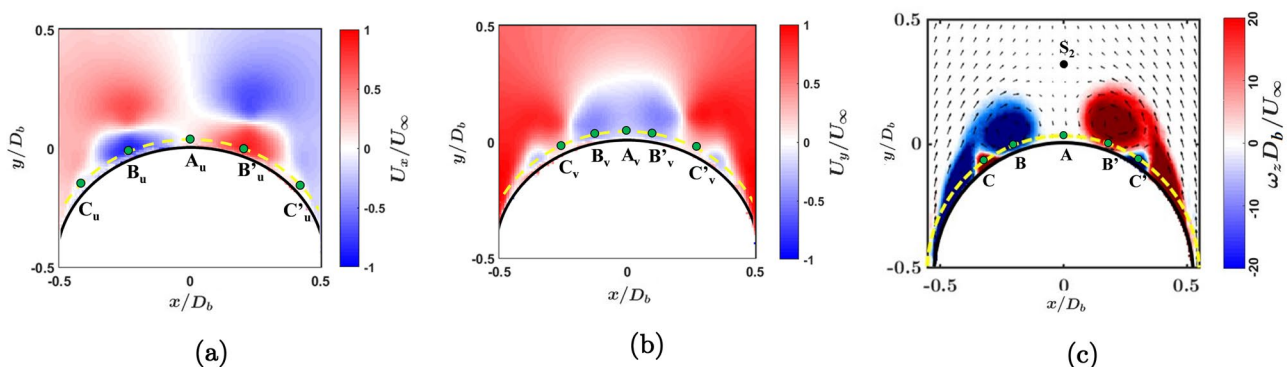


Fig. 22 The contour of extraction is shown for **a** U_x , **b** U_y , and **c** Vorticity ($z/D_b = 2.31$)

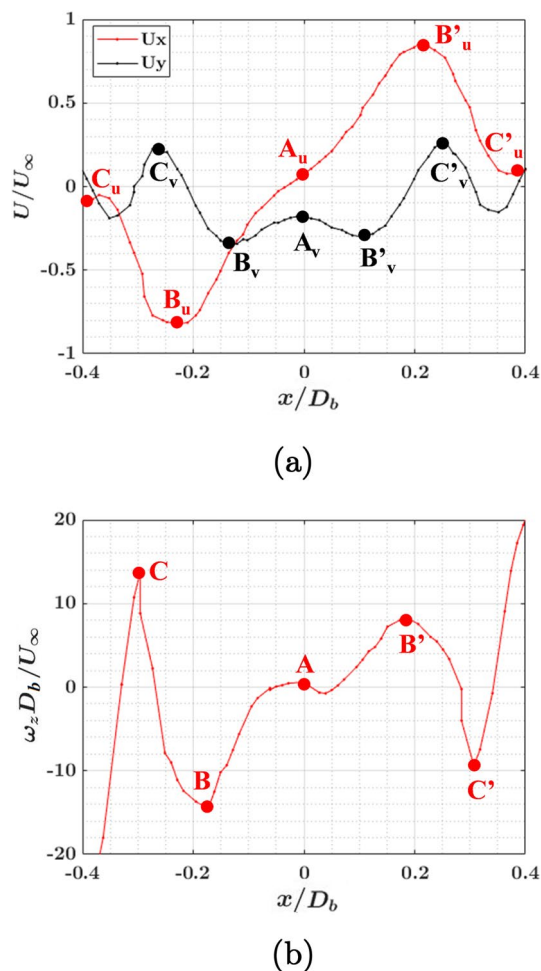


Fig. 23 Contours of **a** velocities and **b** vorticity near the surface for symmetric flow at $z/D_b = 2.31$ for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

and B'_u) around $x/D_b = \pm 0.2$, just before the secondary vortex. Beyond this point, $|U_x|$ decreases to almost zero (C_u and C'_u) in the secondary vortex and the neighboring shear layer. This decrease may be primarily because the contour of the surface and the adjoining flow is gradually bending towards the y-axis. U_x is positive on the starboard-side and negative on the port-side, as is observable from the flow-field. A change of sign is not captured near the secondary vortex region. This is probably because the secondary vortex is very small, and its center of circulation is beneath the contour of extraction. The U_y profile is nearly symmetric with respect to $x/D_b = 0$, unlike U_x . The flow is in the negative y-direction as it moves from the saddle point S_2 towards the body. Therefore, U_y is negative below S_2 at $x/D_b = 0$ (indicated by the point A_v). It continues to become more negative, reaching a minima $\approx |0.3|$ (B_v and B'_v) around $x/D_b = \pm 0.15$, just before the secondary vortex, on both port and starboard sides. This is followed by positive peaks (C_v and

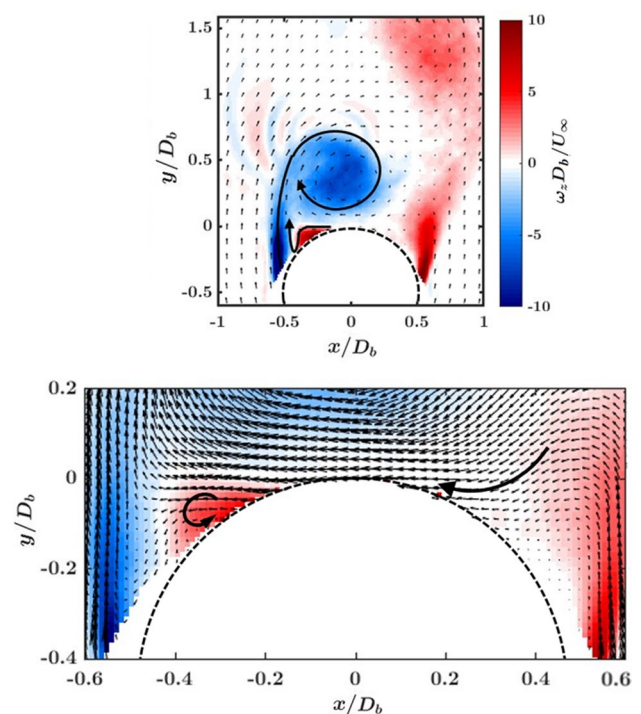


Fig. 24 Characteristics of asymmetric flow over the slender body at $z/D_b = 7.6$ for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

$C'_v \approx 0.2$) around $x/D_b = \pm 0.25$, caused by the movement of flow around the secondary vortex.

The vorticity (ω_z) profile is asymmetric with respect to $x/D_b = 0$, like U_x . It is zero below the saddle point (indicated by the point A in the plot). On the port-side, it shows a minima ($B \approx -14$) corresponding to the primary vortex and a maxima ($C \approx 14$) corresponding to the secondary vortex. It becomes negative as the contour enters the shear layer. The starboard side shows similar variation but of opposite sign. Differences are observed in the peak levels of both velocity and vorticity, suggesting that the flow is slightly asymmetric.

3.3.2 Highly asymmetric vorticity field

Figure 24 shows the highly asymmetric flow-field near the end of the cylinder and the zoomed-in view over the surface, constituting the second characteristic flow. The port-side vortex is prominently placed over the center of the body while the starboard-side vortex has lifted away from the surface. We observe that on the port-side, the flow along the surface separates to form the secondary vortex. The flow attaches after secondary vortex and interacts with the separated shear layer. A clear elliptical vortex is not observed as a result of this interaction, unlike in the symmetric case, Fig. 21. Due to the placement and size of the port-side vortex, the flow is moving from the top of the cylinder to its side on the port-side. While on the starboard side, it is moving

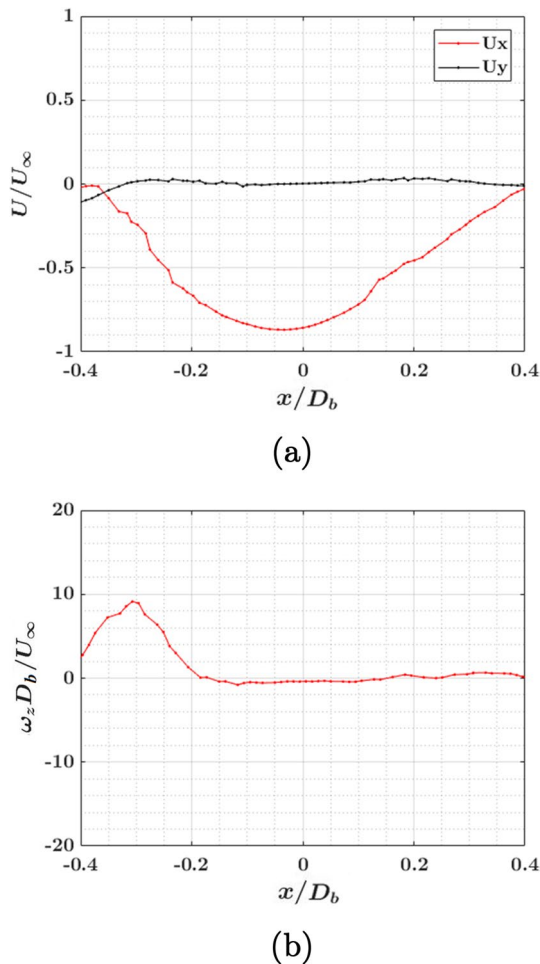


Fig. 25 Contours of **a** velocities and **b** vorticity near the surface for asymmetric flow at $z/D_b = 7.6$ for $\alpha = 40^\circ$ and $\Phi = 180^\circ$

from the side of the cylinder to the top, as indicated by the arrow in the zoomed view. Unlike the port side, a secondary vortex was not found in our measurements on the starboard side.

The velocity and vorticity profiles extracted near the surface are shown in Fig. 25. The U_x profile is completely different from the symmetric case. It lacks the minima and the maxima with a stagnation point at $x/D_b = 0$, as seen in Fig. 23. Instead, it is negative over the entire region with a minima located near the top of the cylinder, $x/D_b \approx 0$. This is because the flow is completely dominated by a single port-side vortex. This will lead to a significant variation in not only the cross-sectional side force but also the normal force since the stagnation point is absent. The magnitude, $|U_x|$, then gradually reaches zero towards the starboard side. The variation in U_y is much smaller in comparison to U_x . The twin-peak profile observed for the symmetric case is absent. It is almost zero everywhere except for a small negative region on the port-side due to the secondary vortex. The vorticity is zero on the starboard side before it reaches the

shear layer. On the port side, it shows a positive peak around $x/D_b = -0.3$ corresponding to the secondary vortex.

3.3.3 Merging of secondary and primary vortices

The third characteristic flow consists of the merging of the primary vortex and the secondary shear-layer vortices. Four main observations are outlined in this discussion. Figure 26 consists of selected planes, showing the development and merging of the first and third sets of the secondary shear-layer vortices. The first set forms at the cone–cylinder junction ($z/D_b = 2.31$) and the third set develops at $z/D_b = 3.6$. The velocity and vorticity contours extracted near the surface, for selected cases are shown in Fig. 27.

The first observation is the difference in the merging characteristics of the first and the third set. For the first set, a secondary shear-layer vortex develops on each side of the cone–cylinder junction. They move up along the contour of the cylinder until they interact with the secondary vortex. During this process, the vortices show a distinct center of circulation. At location $z/D_b = 2.7$, the secondary shear-layer vortices (elliptic in shape) can be identified adjacent to the secondary vortex. However, planes $z/D_b = 3.1$ and $z/D_b = 3.4$ show that as soon as these secondary shear-layer vortices leave the cylinder surface, a distinct center of circulation is no longer visible. They seemed to be stretched into shear layers and continue to be so for the rest of the merging process. Plane $z/D_b = 3.4$ also shows the second set of secondary shear-layer vortices moving close to their respective secondary vortex, while the first set is still in the merging process. The second set shows very similar characteristics to the first and hence not presented in the discussion. The third secondary shear-layer vortex only develops prominently on the port side, locations $z/D_b = 4.7$ – 5.7 . It develops from the surface imperfections at the location where the force balance is attached to the model. The imperfections are uncontrolled and so are their interactions with the freestream, which might explain the formation of the vortex on only one side. Unlike the first two sets, the merging of the third set is more resembling of the merging of two equal-strength co-rotating vortices. At plane $z/D_b = 5.2$, the primary and the secondary shear-layer vortices are in the merging process with two distinct centers of circulation. At plane $z/D_b = 5.7$, the vortices have almost merged, with a single center of circulation observable at the vorticity centroid. Hence, two different merging mechanisms are observed over the slender body.

The second observation is that since the third secondary shear-layer vortex is only formed on the port-side, it leads to significant observable asymmetry in the field. The merged port-side vortex at plane $z/D_b = 5.2$ has crossed the centerline of the cylinder pushing the starboard side vortex away, a phenomenon not observed in the first set of the secondary shear-layer vortex merging process. The same point

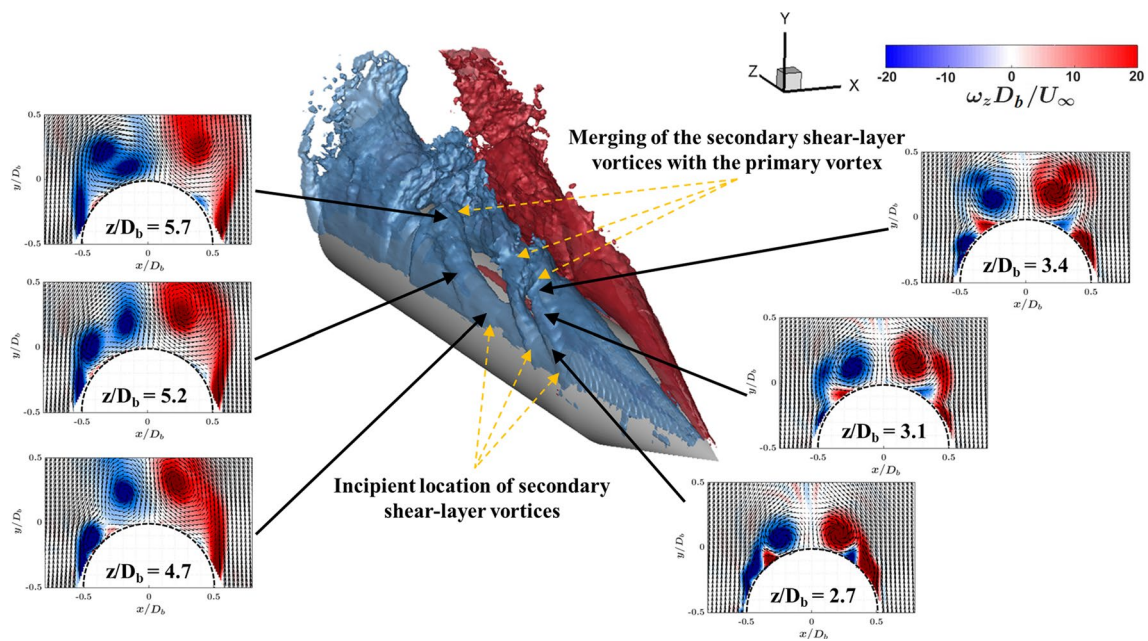


Fig. 26 Development and merging of secondary shear-layer vortices

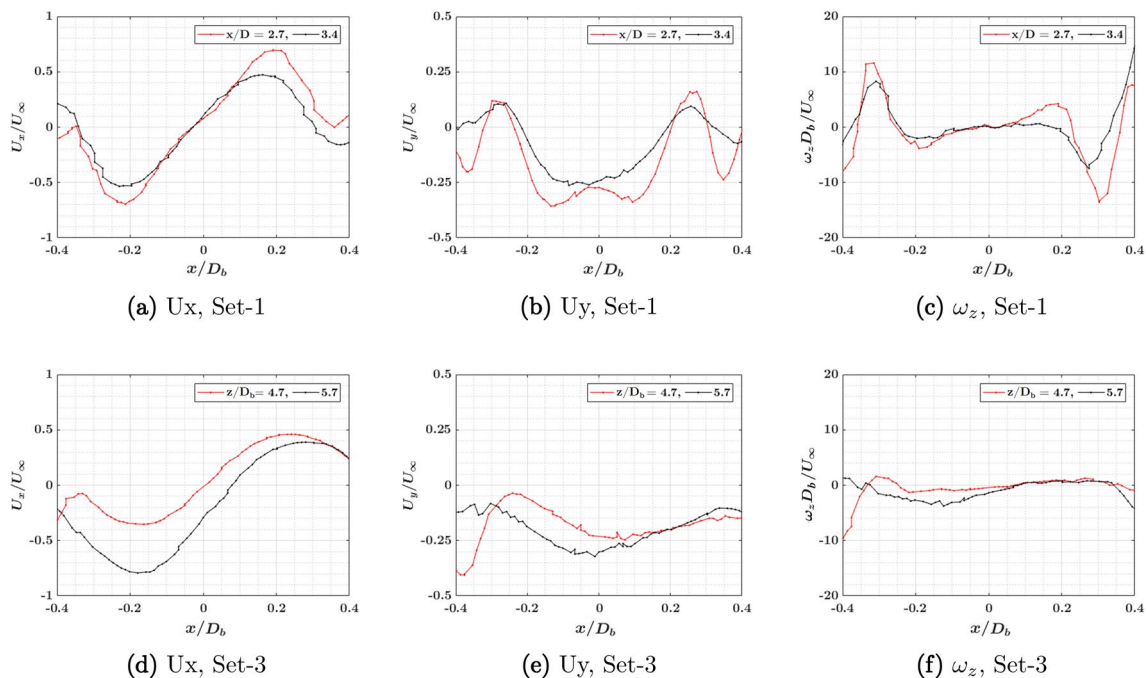


Fig. 27 Contours of velocities (**a**, **b**, **d** and **e**) and vorticity (**c** and **f**) near the surface during the merging of secondary shear-layer vortices

emerges in comparing the U_x velocity profiles in Figs. 27a, d. These figures compare the profiles corresponding to the first and the last plane from the first set and the third set of secondary shear-layer vortices. The profiles have a positive and a negative region corresponding to the surface below the starboard and port-side vortex, respectively, very similar to

the symmetric field observed in Fig. 21. The stagnation point between the two vortices for the first set is located almost at the centerline ($x/D_b \approx 0$), while for the third set, it has clearly moved away from it (to $x/D_b \approx 0.1$). A high level of asymmetry will be expected to produce a large sectional side-force. This also suggests that a mechanism by which

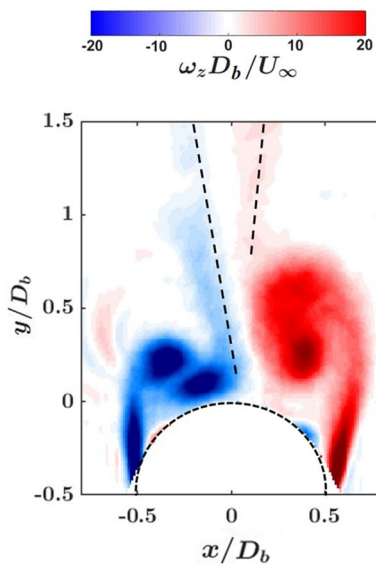


Fig. 28 Vorticity filaments developed during merging process ($z/D_b = 5.7$)

the cylindrical aftbody can produce significant side-force is when a secondary shear-layer vortex only develops on one side of the body.

The third observation is that the characteristics of the secondary vortex vary significantly during the interaction with the secondary shear-layer vortices. Visually, the shape of this secondary vortex can be observed to be varying for the first set and significantly diminished for the third set of secondary shear-layer vortex merging. A stark difference is observed on comparing the U_y and ω_z profiles corresponding to Set-3 (Fig. 27e, f) and Set-1 (Fig. 27b, c). The two positive peaks (around $x/D_b = \pm 0.3$) in the U_y profiles of Set-1, Fig. 27b, are due to the rotation of the flow about the secondary vortex. The peaks for Set-3, Fig. 27e, are significantly reduced and do not attain a positive value. The secondary vortices being very small, fail to produce a significant variation on the U_y component. The vorticity profiles for Set-1 also show two peaks at $x/D_b = \pm 0.3$, corresponding to the two secondary vortices, Fig. 27c. Figure 27f shows that the peaks are almost negligible for Set-3.

The fourth observation is the appearance of longitudinal vorticity filaments, highlighted using dotted lines in Fig. 28. The first and second sets of secondary shear-layer vortices get stretched into shear layers, which rotate in the vorticity fields of the primary vortex-pair before merging with them. The picture emerges clearly in the port-side vorticity iso-surfaces of Fig. 7. It is this shear layer that appears as filaments in the cross-sectional vorticity field. The pair of filaments observed in Fig. 28 were generated from the stretching of the second set of secondary shear-layer vortices.

To summarize, the flow-field over the conical forebody consists of the nearly symmetric primary and secondary counter-rotating vortex pairs. Whereas, on the cylindrical aftbody, secondary shear-layer vortices develop and merge with the existing flow-field, creating significant flow asymmetry. In the present study, the secondary shear-layer vortices seem to enhance the vortex asymmetry of the primary vortex, whereas there can be a case of secondary vortices leading to the suppression of vortex asymmetry. While the imperfections in this study were uncontrolled, the results suggest that controlled surface perturbations on the aftbody can be used to generate secondary vorticity to enhance or suppress vortex asymmetry.

4 Conclusions

The primary objective of the present experimental investigation was to study the initiation and development of asymmetric vortices over a long cone–cylinder body at high angles of incidence. The present study emphasizes the importance of joints and surface imperfections on the flow field so that other researchers and the scientific community at large are aware of such features, particularly those studying problems involving high angles of incidence. Force measurements showed that rotation of the conical forebody while keeping the cylinder roll orientation fixed, led to a periodic variation in the net side-force coefficient at high angles of incidence. However, the coefficient did not switch from positive to negative values during the periodic variation, which is generally observed for just a conical forebody. This observation is attributed to the fixed roll orientation of the cylinder and the imperfections on its surface.

PIV measurements showed that a pair of counter-rotating vortex pair initiates at the tip of the cone and grows over the length of the body through the assimilation of the separated shear layer. The development of the primary vortex pair on the highly polished cone is nearly symmetric, with little observable variation with the roll. However, the vortices become asymmetric while developing over the cylinder. The comparative strengths, and locations of the two primary vortices with respect to the body-axis, vary as they grow along the cone–cylinder body.

The development of asymmetry starts at the cone–cylinder junction and seems to vary with the roll orientation of the test model. The separated shear layer from the cylinder forms secondary shear-layer vortices at several locations along the length of the body, including the cone–cylinder junction. The characteristics and number of secondary vortices formed differs on the two sides of the cylinder. These secondary shear-layer vortices generated due to surface imperfections and joints, merge with the primary vortex pair originated from the cone-tip, leading to flow asymmetry. The

greatest extent of asymmetry was observed for the roll position corresponding to the highest side-force coefficient, indicating that side forces observed at high angles of incidence are due to vortex asymmetry of the flow-field. At this roll position, the vortex trajectories were also highly asymmetric with respect to the axis of the body. Near the trailing edge, one of the vortices encompassed most of the region over the cylinder while the other had moved far away from the body. These results suggest that cone–cylinder junction and surface imperfections on the cylinder play a critical role, in addition to the surface roughness of the conical tip, towards the development of vortex asymmetry.

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